

PART-B

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Signature of the Invigilator: Dayan

INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-B of the examination has **4 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No questions require any clarification from the instructor or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	4	Total
Points:	6	7	5	2	20
Score:	NA	5	5	1	11

QUESTIONS

1. (6 points) Let

$$S = \begin{pmatrix} S_{yy} & S_{yX}^T \\ S_{yX} & S_{XX} \end{pmatrix}$$

be the partitioned sample covariance matrix of $Y_{1 \times 1}$ and $X_{q \times 1}$ based on a centered (with respect to sample mean) sample of size n . Prove that the squared sample first canonical correlation r_1^2 is identical to the coefficient of determination (R^2) obtained from the multiple linear regression of Y on the vector X .

Ans :

7

[Faint handwritten text, likely bleed-through from the reverse side of the page. The text is mostly illegible due to fading.]

2. Suppose you are performing a factor analysis on a 4×4 sample covariance matrix $\mathbf{S}$. The eigenvalues of $\mathbf{S}$ are $\lambda_1 = 0.8$, $\lambda_2 = 2.1$, $\lambda_3 = 4.2$, and $\lambda_4 = 0.2$. The corresponding normalized eigenvectors are $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, and $\mathbf{e}_4$. You decide to extract $m = 2$ factors using the principal component method.
- (2 points) Write down the explicit mathematical expression for the estimated factor loading matrix $\tilde{\mathbf{L}}$ in terms of the given eigenvalues and eigenvectors.
 - (1 point) Let the first row of $\tilde{\mathbf{L}}$ be $\tilde{\mathbf{l}}_1^T = [0.8, -0.4]$. Calculate the estimated communality $\tilde{h}_1^2$ for the first variable.
 - (4 points) Prove that the sum of all estimated communalities $\sum_{i=1}^p \tilde{h}_i^2$ is exactly equal to the sum of the largest $m = 2$ eigenvalues of $\mathbf{S}$.

Ans:

$$(a) \tilde{\mathbf{L}} = \begin{bmatrix} \sqrt{\lambda_1}(\mathbf{e}_1) & \sqrt{\lambda_2}(\mathbf{e}_2) & \sqrt{\lambda_3}(\mathbf{e}_3) & \sqrt{\lambda_4}(\mathbf{e}_4) \end{bmatrix}$$

$$(b) \tilde{h}_1^2 = (0.8)^2 + (-0.4)^2 \quad (\text{extracting } m=2 \text{ factors})$$

$$= 0.64 + 0.16$$

01 $\tilde{h}_1^2 = 0.80 \rightarrow$ Estimated communality for 1st variable.

$$(c) \sum_{i=1}^p \tilde{h}_i^2 = \sum_{j=1}^n (L_{1j}^2 + L_{2j}^2) \quad \text{extracting } m=2 \text{ factors (PCA).}$$

$$= \sum_{j=1}^n L_{1j}^2 + \sum_{j=1}^n L_{2j}^2$$

$$= \sum_{j=1}^n (\sqrt{\lambda_1} e_{1j})^2 + \sum_{j=1}^n (\sqrt{\lambda_2} e_{2j})^2$$

$$= \lambda_1 \left(\sum_{j=1}^n e_{1j}^2 \right) + \lambda_2 \left(\sum_{j=1}^n e_{2j}^2 \right)$$

L_{1j}, L_{2j} correspond to ~~the~~ the greatest two eigenvalues λ_1, λ_2 .

$\lambda_1 \geq \lambda_2 \geq \dots$
Order of eigen values.

04 Here, the eigenvectors are normalized. So, for all $1 \leq i \leq p$, $\sum_{j=1}^n e_{ij}^2 = 1$.

$$\rightarrow = \lambda_1(1) + \lambda_2(1)$$

$$= \lambda_1 + \lambda_2 = \text{Sum of largest } m \text{ eigen values}$$

Let $f: X \rightarrow Y$ be a function and $A \subseteq X$.

Prove that $f(A) \subseteq f(X)$.

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3. (5 points) Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a distribution with mean $\mu \in \mathbb{R}$ and variance μ^2 . Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Derive a variance-stabilizing transformation $g(\bar{X}_n)$ such that the asymptotic variance of $g(\bar{X}_n)$ does not depend on μ .

Ans: $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \rightarrow$ mean of Random sample

from CLT (Central Limit Theorem), for any distribution X_i ,

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0, 1) \quad \text{as } n \rightarrow \infty$$

here, $\sigma^2 = \mu^2$

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\mu} \xrightarrow{d} N(0, 1)$$

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \mu^2) \quad (1)$$

Applying Taylor's theorem for $g(\bar{X}_n)$,

$$g(\bar{X}_n) \approx g(\mu) + (\bar{X}_n - \mu) g'(\tilde{\mu})$$

$$\tilde{\mu} \in (\mu, \bar{X}_n) \quad \text{if } \mu < \bar{X}_n$$

$$\sqrt{n}(g(\bar{X}_n) - g(\mu)) \approx g'(\tilde{\mu}) \sqrt{n}(\bar{X}_n - \mu)$$

$$\tilde{\mu} \in (\bar{X}_n, \mu) \quad \text{if } \mu > \bar{X}_n$$

$$\text{as } n \rightarrow \infty, \sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \mu^2)$$

$$g'(\tilde{\mu}) \rightarrow g'(\mu)$$

from Slutsky's theorem, if $X_n \rightarrow X$ and $Y_n \rightarrow c$, $X_n \cdot Y_n \rightarrow c \cdot X$

$$\text{So, } \sqrt{n}(\bar{X}_n - \mu) g'(\tilde{\mu}) \xrightarrow{d} g'(\mu) N(0, \mu^2) = N(0, (\mu g'(\mu))^2)$$

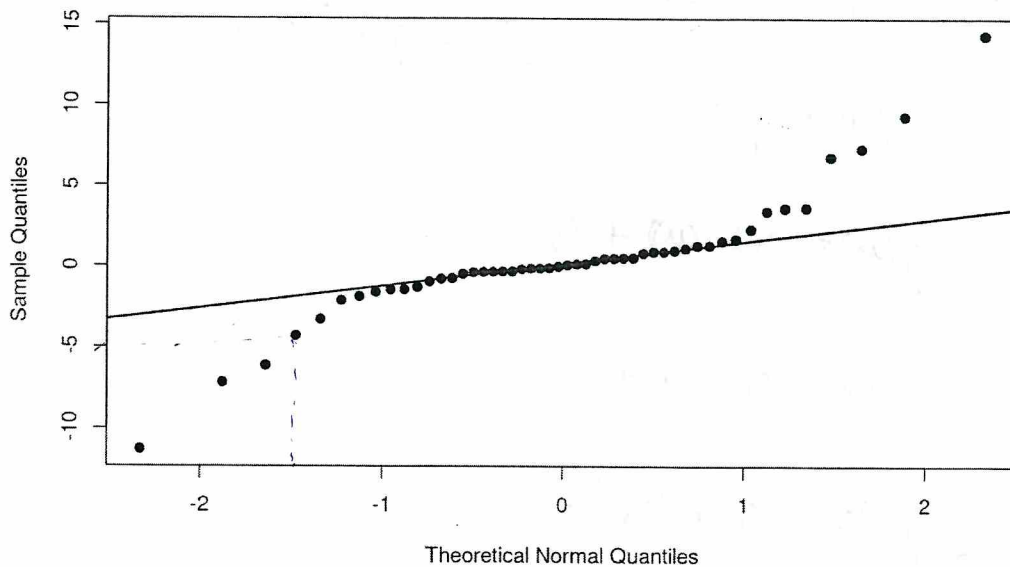
$$\sqrt{n}(g(\bar{X}_n) - g(\mu)) = \sqrt{n}(\bar{X}_n - \mu) g'(\tilde{\mu}) \xrightarrow{d} N(0, (\mu g'(\mu))^2)$$

Asymptotic variance $= (\mu g'(\mu))^2 \rightarrow$ for this to not depend on μ ,

~~for~~

[[Continued in Page 8.]]

4. (2 points) The figure below displays a Normal Q-Q plot for a dataset of size $n = 50$. Interpret this plot to determine if the data follows a normal distribution. If the normality assumption is violated, specify which class of distributions would more accurately characterize the data.



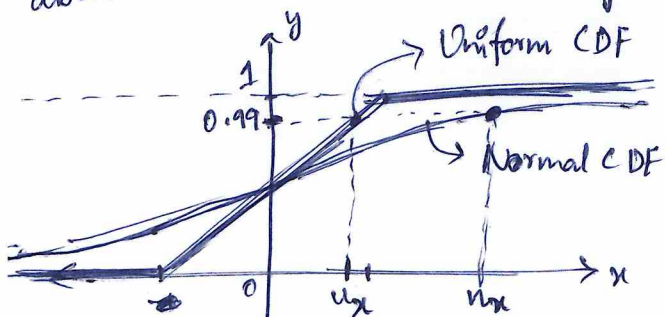
Ans: This Q-Q plot data would ~~be more~~ be characterized by Uniform distribution more accurately.

Because, plot follows an inverted S-shape, showing most density in the middle region but also few outliers in the left and right regions, showing

it may not be a normal distribution.

~~After~~ ~~Before~~ ~~is~~ ~~not~~

In the middle, uniform distribution is close to Normal distribution, which is observed in the plot. At far away negative points, ~~suppose~~ CDF of uniform distribution is ~~above~~ below normal and at far positive points, CDF of uniform is above normal, which is clearly observed in the plot.



Ex: at a far positive point :
for CDF value = 0.99, Uniform distribution reaches it in lesser x than normal distribution with clear difference.

[Q₃ continued... $(c_1 > 0)$

$$\text{Let } u g'(u) = c_1 \iff \text{constant} \implies (u g'(u))^2 = c_1^2$$

$$g'(u) = \frac{c_1}{u}$$

$$g(u) = c_1 \ln(u) + c_2$$

$$\text{So, } g(u) = c_1 \ln u + c_2$$

$$g(\bar{X}_n) = c_1 \ln(\bar{X}_n) + c_2$$

For this transformation g ,

05

Asymptotic variance $= c_1^2 \implies$ does not depend on u .

